

TJWA Nipping / Laminating Best Practices

Safety and Waste	
1. Eliminate unnecessary nips from your process.	Nips are hazardous - Dangerous to life and limb. Nips create waste from wrinkling (especially with baggy web) and lateral shifting. Some nipped rollers have high wear rates and short lives, creating process variation and ongoing regrinding and recovering costs.
2. Safety! Follow nip guarding rules.	Nips are hazardous to your health. Properly guard required nip points.
How Much Nip?	
3. Load or pressure controlled nips are preferred to gapped or indentation controlled nips.	Load and pressure controlled nips will float with product thickness variations. Gap or indentation controlled nip pressures will be dependent on product thickness and compressibility.
4. Run most nipped processes to a target footprint pressure.	Material respond to stress or force over area. Many polymer processes are time, temperature, pressure functions. The nip provides two of these critical variables (time and pressure). To get nip footprint pressure, you have to measure it directly or follow these recommended steps to estimate it.
5. Calculate the nip force from the combination of gravity (roller weight) and external load.	Nips closing horizontally will often be independent of roller weight. Nips closing vertically from above will add roller weight to the nip force. Nips closing vertically from below will subtract roller weight from the nip force. Geometries at angles other than vertical or horizontal will require trigonometry to determine the nip force contribution of roller weight.

6. Determine effective the effective nip width is the web or roller width, w_{eff} .	If the web is thick relative to roller conformance (as in most steel on steel nips), the effective nip width is the web width. If the web is thin relative to the roller conformance, then the roller defines the nip width.
7. Calculate nip load per width, $N = F(\text{nip})/w_{\text{eff}}$, from nip force and effective width or from roller indentation.	As an alternate to estimating nip force and effective width, rubber nip loads can be estimated from roller geometry, rubber hardness, and indentation. (Indentation can be estimated from footprint length.)
8. Roller width should be greater than web width, but not excessively wider than the web (e.g. roller width is < 105% of web width)	Running a narrow web on wide nip rollers can create uneven wear and may mean most of nip load is made outside the web in roller-to-roller contact. Most webs will be damaged by nip roller contact less than web width. Damage may be wrinkling (from the shear of the transition from contact to no contact) or impressions from the 'sharp' roller edge.
9. Create a table or chart of N or $F(\text{nip})$ vs. supply pressure.	Help your operators to understand the load in units of force created by the nipped system. It can be hard to appreciate how 50psi (3bar) can create 2000 lbs load (1000 kgf).
10. Measure the footprint and TD footprint variations, b , checking for uniformity vs width. Target less than 10% variations of b vs width.	Footprint is an easy measurement and a good indicator of crossweb pressure. Footprint can be measured in many ways, from the simple tape bubble test to devices costing thousands of dollars. Dynamic footprint devices avoid false center measurements of high deflection 'open-center' nips.
11. Estimate process pressure from load per width and footprint $P=N/b$	Average pressure is force per width per machine direction contact length (footprint length). Hertzian pressure will be parabolic in the machine direction rising and falling from zero pressure, with a maximum pressure of
12. For gapped or negative gapped nips, measure the gap.	Many gapped or indentation controlled nips have micrometers to adjust and measure the critical dimension.

How To Nip?	
13. Pivoting nips are preferred to linear sliding nips.	Pivoting nip systems are easier to keep aligned and less prone to sticking or high friction.
14. Load should be delivered via pneumatic or hydraulic pressure.	Actual loads are difficult to determine with spring or rubber compression loading.
15. Both rigid and articulating nipping systems can be successful.	Articulating nips (where one roller is square and second roller flexes in the closing direction to roller and product variations) are best for pressure controlling processes (lamination, winding). Articulating nips float to roller and web imperfections, allowing more uniform pressure profile. Rigid, square nipping of rollers is preferred in thickness controlling processes (coating, calendaring).
16. Use flow control to prevent nips from slamming shut or flying open.	Controlled opening and closing is safer and will avoid shock-induced misalignment. For articulating nips, try to adjust flow so both end contact simultaneously. Opening should be quick, for safety, but not unrestricted.
17. Nips should open when process is stopped for more than a few minutes.	Rubber is elastic under short time periods but will form a flat spot is left closed for long periods.
18. Nip point should be well-lit and easily viewable.	Nip points are a common source of wrinkling and must to seen to be prevented.
19. Webs should run on the nip centerline.	Offset webs in nips will see more nip-induced lateral tracking, wrinkling, and pressure variations.

20. Avoid wedging nips.	Nip load should close nearly parallel to a line draw between the roller axes. Wedging nips can create loads much higher than anticipated, even high enough to damage rollers.
Compliance and Deflection	
21. Two hard rollers may be used when the product is compliant.	For compliant materials (non-wovens, textiles, papers, coatings), the nipping process goal is often to compress or meter the compliant material to uniform thickness profile.
22. At least one compliant roller is preferred for pressure-controlled processes.	Roller compliance conforms around thickness and diameter variation, reduce pressure variations (but lowers the pressure for a give load).
23. Rule of Thumb: Roller length/diameter < 5. Goal: Keep roller deflection under process conditions to be less than 2-6 mils (50-150 microns), or less than 20% of average indentation.	This guideline for 2-6 mils (50-150 microns) of deflection is based on high modulus film, paper, and foil handling, considering both process sensitivity and equipment cost. Precision nipped processes or foils handling may require better than this goal. Thicker, low modulus, or more compliant web may relax this specification.
24. Rubber rollers should have 10-30 mils (0.25-0.75mm) indentation under anticipated process pressures (so that deflection causes less than 20% process variation).	If the deflection goal or process requirements are tighter or looser than 2-6 mils (50-150 microns) then this indentation specification can be adjusted, accordingly.
25. Default should be cylindrical roller. Cylindrical rollers should have diameter variations less than 10% of indentation.	Advanced rubber roller suppliers have scanning laser systems to confirm roller diameter profile and should have less than +/- 2 mil (+/- 50micron) variation.

26. Crowned rollers are effective, especially when compliant rollers are not an option (usually due to high temperature processing).	Crowned rollers can be effective at countering pressure variations from roller deflection for a given roller combination and specific load. Crowning is usually used in system where either high temperature or high load makes rubber covered roller impossible or short-lived. Crowning will create more variation (low-high-low pressure profile) if used greatly below their design load.
27. Nips should open when process is stopped for more than a few minutes.	Rubber is elastic under short time periods, but will form a flat spot is left closed for long periods.
Tensions and Curl	
28. Minimize tension-induced curl by tensioning each web to match pre-laminating strains.	Webs are usually combined under the machine direction strain of tensioning. When tension is removed, the elastic webs will return to their unstrained dimensions. If the strains are mismatched at laminating, the untensioned laminate will have uneven lengths, creating curl toward the shortest layers.
29. Consider strains from thermal and hygroscopic expansion coefficients.	Strain mismatches can be created by dimensional changes other than from tension. Papers and BOPA (Nylon) films will expand and contract with moisture content. Nearly all materials will grow and shrink with temperature changes.
30. Consider Poisson's ratio of a laminate's layers. If they are mismatched, laminate at lowest tension possible to avoid TD curl.	If a laminate's Poisson's ratio are greatly different (a.k.a. laminating a non-woven to a film), then matching strains in the machine direction will create mismatched transverse direction strains and TD curl.
31. Avoid post-nip wrap on nipping rollers.	In some products, wrapping after the nip will induce curl in the direction of post-nip wrap. The mechanism of this curl is not clear, but the geometry fo the first milli-seconds of contact influences final shape.
32. Set exit tension equal to the sum of the input tensions.	

33. Most nipped roller systems will have a driven steel roller and an idling or torque driven rubber roller.	The steel roller will have an rpm to surface speed independent of nip load.
34. If needed, drive the compliant roller with torque control.	For some products, such as shear sensitive thick, soft adhesives or substrates, driving the rubber roller prevents product damage.
Wrinkling and Delaminating	
35. Entry rollers for each web should create short spans and small pre-nip point wrap angles.	Short entry spans are stiff and will reduce lateral shifting, center gathering, and related wrinkling better than long entry spans. A small amount of pre-nip point wrap angle can provide some web-roller traction to reduce in-nip speed variations from shifting or gathering the entering web.
36. Avoid pre-nip wrinkling with a spreader immediately upstream of the nip point.	Uneven nipping (from diameter, loading, and web variations) can be reduce by spreading the web to lateral tautness as close as possible to the nip entry point. Spreading immediately before nipping will also help pull out web bagginess and associated problems.
37. Avoid 90-degree wraps on nipping rollers to prevent deflection-induced tracking wrinkles.	End-supported nip rollers will tend to deflection away from each other. A web entering parallel to this deflection will tend to track towards centerline, promoting wrinkling.
38. Minimize accumulation wrinkles by pulling out bagginess with tension.	Tension will elongate the short lanes of web and with enough tension, will make baggy web planar with equal transverse position length, allowing metering through uniformly pulling nipped roller without wrinkling.
39. Avoid larger wrap angles on all rubber nip rollers.	Compliant covering compress and elongate in the nip point, creating a speed differential between the surface speed of the nipped and unnipped rubber. Stiff webs will not elongate as much as a rubber covering, creating microslip in the wrapped contact. More wrap angle and more micro-slip contact will create faster roller (or web) abrasion and debris.

40. Use large wrap angles on steel nip rollers only if needed for pre-heating.	Steel rollers do not have speed variations between nipped and unnipped contact, making micro-slip a non-factor. However, steel rollers can still have deflection problem (avoid 90-degree pre-nip wrap) and benefit from small pre-nip traction (avoid 0-degree pre-nip wrap).
41. For cambered webs, use a pre-nip skewing roller to avoid wrinkles.	If a web has high camber (an asymmetric length difference not 'pulled out' with tension), a skewing roller immediately upstream of a nip point can temporarily tighten the loose edge and prevent accumulation wrinkles. A skewing roller is best when misaligning parallel to the web span into the nip roller and with close to 90-degree wrap to reduce wrinkles or lateral shifting from the misalignment.
42. Avoid or minimize wrapping crowned rollers with wide webs.	As webs with good traction will shift toward larger diameter, a crowned roller will promote strong centering from both sides of a web, often forming a central wrinkle and crease in a web.
43. To avoid delamination after laminating, avoid small diameter rollers in handling thick laminate with weak green strength.	Bending over a roller creates strain changes and shear proportional to the thickness divided by the radius. Thick webs on small radius rollers can curl from yielding or delaminate from shear.
44. Use laterally ridged rollers to prevent defects associated with build up ahead of rollers.	A laterally ridged roller (usually with some angle or chevron shape) can 'pump' air or delamination over a roller in small amounts and prevent the tunneling defect of a trapped pre-roller bubble or wrinkle or a large pre-roller delamination.